



Voting systems

Task A: Single Transferable Vote (STV)

1) To decide the quota required to be elected, the following formula is used:

$$\frac{\text{total votes}}{\text{total seats} + 1} + 1$$

Calculate the quota for the following elections:

- a) 12 574 votes cast and three seats to be filled. 3145 votes
- b) 12 574 votes cast and four seats to be filled. 2516 votes
- c) 12 574 votes cast and five seats to be filled. 2097 votes

2) What do you notice about the number of votes required?

As the number of seats available increases, the number of votes required to reach the quota (get elected) decreases.

3) 20 000 votes are cast in an election. There are three seats available. Here is a table showing the breakdown of the votes.

First vote	Second vote			
A: 5813	B: 462	C: 1068	D: 1767	E: 2516
B: 7820	A: 1564	C: 2737	D: 539	E: 2980
C: 1292	A: 259	B: 543	D: 87	E: 403
D: 3983	A: 112	B: 92	C: 2350	E: 1429
E: 1092	A: 415	B: 317	C: 268	D: 92

a) Calculate the quota required to win a seat.

5001

$$\frac{\text{total votes}}{\text{total seats} + 1} + 1$$

b) Based on the first vote, who has already won a seat?

Candidates A and B have both won seats as their vote total is over 5001

c) Assign all excess votes from any successful candidate(s) to the candidate(s) who were the most popular second vote. $E = 4723$

d) Who has won the final seat?

Candidate E has won the final seat as C would now be eliminated.



Task B: Alternative Vote (AV)

1) 20 000 votes are cast in an election. Here is a table showing the breakdown of the votes.

First vote	Second vote			
A: 5813	B: 462	C: 1068	D: 1767	E: 2516
B: 7820	A: 1564	C: 2737	D: 539	E: 2980
C: 1292	A: 259	B: 543	D: 87	E: 403
D: 3983	A: 112	B: 92	C: 2350	E: 1429
E: 1092	A: 415	B: 317	C: 268	D: 92

a) How many votes are required to win the seat?

Over 10 000 (10 001 or more votes.)

b) Since no one has over half the vote, which candidate is eliminated?

Candidate E

c) Assign the votes accordingly and produce an updated table.

Vote count
A: 6228
B: 8137
C: 1560
D: 4075

d) Which candidate(s) can win the election? Justify your answer.

In the next round, candidate C will be eliminated.

Even if all their votes go to candidate D, candidate D would only have 5635 votes, which means they would be eliminated next.

Depending on where the votes from eliminated candidates go, either A or B could still win.

